

```
public class BinarySearch {  
    public static int binarySearch(int[] arr, int target) {  
        int left = 0;  
        int right = arr.length - 1;  
  
        while (left <= right) {  
            int mid = left + (right - left) / 2;  
  
            if (arr[mid] == target) {  
                return mid; // Target found  
            }  
            if (arr[mid] < target) {  
                left = mid + 1; // Search right half  
            } else {  
                right = mid - 1; // Search left half  
            }  
        }  
        return -1; // Target not found  
    }  
  
    public static void main(String[] args) {  
        int[] sortedArray = {2, 5, 8, 12, 16, 23, 38, 45, 56, 72};  
        int target = 23;  
        int result = binarySearch(sortedArray, target);  
  
        if (result != -1) {
```

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=== Code Execution Successful ===

